Hoja # 4

1) Datos:
$m = 18 kg$
$h = 12 m$
$g = 9.8 m/s^2$

a) $E_T = mgh = 18 \cdot 9.8 \cdot 12 = 2116.8 J$

$E_c = 0 J$

$E_T = 2116.8 J$

b) A 3m sobre el suelo

$E_p = 18 \cdot 9.8 \cdot 3 = 529.2 J$

$E_c = T_T - E_p = 2116.8 - 529.2 = 1587.6 J$

$E_T = 2116.8 J$

c) En el suelo $h = 0$

$E_p = 0 J$

$E_c = 2116.8 J$

$E_T = 2116.8 J$

2) Datos:
$m = 4 kg$
$h_0 = 10 m$
$h = 4 m$
$g = 9.8 m/s^2$

$E_T = mgh_0 = 4 \cdot 9.8 \cdot 10 = 392 J$

$E_p = 4 \cdot 9.8 \cdot 4 = 156.8 J$

$E_c = E_T - E_p = 392 - 156.8 = 235.2 J$

3) Velocidad martillo Antes de caer

$E_c = \frac{1}{2} m v^2$

$E_c = \frac{1}{2} 4 v^2$

$\frac{392}{2} = v^2$

$392 = \frac{4}{2} v^2 \rightarrow 2v^2$

$v^2 = 196$

$v = \sqrt{196}$

$v = 14 m/s$

4m del suelo

$\frac{1}{2}(4)v^2 = 235.2 J$

$2v^2 = 235.2$

$v^2 = 117.6$

$v = 10.84$

4) Datos
$m = 5\,kg$
$h = 10\,m$
$g = 9.8\,m/s^2$

$\frac{1}{2}mv^2 = mgh$

$\frac{1}{2}(5)v^2 = 5 \cdot 9.8 \cdot 10$

$2.5v^2 = 490$

$v^2 = 196$ $\quad v = \sqrt{196} = 14\,m/s$

5) Datos
$m = 100\,kg$
$h = 45\,m$
$g = 9.8\,m/s^2$

$\cancel{m}gh = \frac{1}{2}\cancel{m}v^2$ $\quad v^2 = 2gh = 2(9.8)(45)$
$= 882$

$gh = \frac{1}{2}v^2$ $\quad v = \sqrt{882} = 29.7\,m/s$

6) Datos
$m = 8\,kg$
$v_A = 7\,m/s$
$h_A = 20\,m$
$h_B = 0$
$g = 9.8\,m/s^2$

$E_A = \frac{1}{2}mv_A^2 + mgh_A$
$E_A = \frac{1}{2}8 \cdot 7^2 + 8 \cdot 9.8 \cdot 20$
$E_A = 196 + 1568$
$E_A = 1764\,J$

$E_B = \frac{1}{2}mv_B^2$ $\quad v_B = 21\,m/s$
$1764 = \frac{1}{2}(8)v_B^2$
$1764 = 4v_B^2$
$v_B^2 = 441$

7) Datos
$h_c = 8\,m$

$E_c = \frac{1}{2}mv_c^2 + mgh_c$

$1764 = \frac{1}{2}(8)v_c^2 + 8(9.8)(8)$
$1764 = 4v_c^2 + 627.2$
$1764 - 627.2 = 4v_c^2$
$1136.8 = 4v_c^2$
$v_c^2 = 284.2$ $\quad v_c = 16.9\,m/s$

8) Datos

$m = 2kg$

$v_0 = 0$

$U_A = 103J$

$k_B = 50J$

$E_A = U_A = 103J$

$E_B = U_B + K_B$

$103 = U_B + 50$

$U_B = 53J$

9) Datos

$U_1 = 107.3J$

$k_1 = 0$

$U = 27.3J$

$E = 107.3J$

$E = k + U$

$107.3 = k + 27.3$

$k = 80J$

10) Datos:

$m = 0kg$

$v_f = 12m/s$

$E_i = mgh$ $\quad mgh = \frac{1}{2}mv^2$

$E_f = \frac{1}{2}mv^2$ $\quad gh = \frac{v^2}{2}$ $\quad h = \frac{v^2}{2g}$

$h = \frac{12^2}{2 \cdot 9.8}$ $\quad h = \frac{144}{19.6}$ $\quad h = 7.35m$

11) Datos

$m = 4kg$

$v = 10m/s$

$E = \frac{1}{2}mv^2$ $\quad E = 200J$

$E = \frac{1}{2}(4)(10^2)$

$E = 2(100)$